CuraTwin

AI-Powered Patient Digital Twin for Glycemic & Cardiovascular Risk

Project Outcomes, Architecture & Clinical Validation

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1. Problem Statement & Clinical Motivation

- The Challenge:
  - 537 million adults live with diabetes globally, facing acute postprandial glucose excursions and a 2x-3x elevated lifetime risk of acute cardiovascular events (MI and stroke).
  - Traditional static risk scores fail to incorporate real-time continuous sensor streams and often suffer from cross-contamination and temporal data leakage in ML training.

- The CuraTwin Solution:
  - An interactive, decoupled patient digital-twin framework unifying wearable CGM telemetry, daily behavioral records, and longitudinal EHR vitals.
  - Serves two independent, specialized ML models trained on strictly partitioned cohorts to prevent leakage:
    1. Model A: 2-Hour Short-Term Glucose Excursion Prediction (>180 mg/dL).
    2. Model B: 365-Day Incident Cardiovascular Event Risk (First Acute MI / Stroke).

2. AI/ML Models & Empirical Validation Outcomes

• Model A (Short-Term Glucose Excursion Forecast):
  - Architecture: XGBoost Classifier (curatwin_glucose_model.joblib)
  - Feature Space: 55 continuous physiological, meal kinetics, and wearable CGM lag features
  - Held-out Disjoint Patient Evaluation: ROC-AUC = 0.9845, PR-AUC = 0.9804, Accuracy = 93.84%, Brier = 0.0470
  - Clinical Utility: Accurately anticipates post-meal glucose spikes exceeding 180 mg/dL 2 hours in advance.

• Model B (365-Day Incident Cardiovascular Event Risk):
  - Architecture: Balanced Random Forest (curatwin_cardiovascular_model.joblib)
  - Feature Space: 38 longitudinal vitals, blood pressure, lipid panels (TG/HDL), and co-morbidities
  - Calibration: Youden's Index optimal screening threshold = 0.0403 (Sensitivity = 83.33%, Specificity = 66.95%)
  - Clinical Utility: Flags high-risk patients for early lipid/antihypertensive optimization.

3. Technical Architecture, Testing & Public Deliverables

• Full Technical Stack:
  - Frontend: Vanilla HTML5, Clean CSS3, Modular JavaScript (SensorAdapter, PatientStateManager)
  - Backend: Python 3.13, FastAPI, Uvicorn, Pydantic v2
  - ML Stack: XGBoost 2.1+, scikit-learn 1.6+, LightGBM 4.5+, Joblib, NumPy, Pandas

• Automated Test Suite:
  - 18/18 API integration tests passed (`tests/test_api.py`)
  - 11/11 Dataset pipeline & zero-leakage tests passed (`dataset/test_pipeline.py`)

• Compliance & Open Access:
  - Open-Source License: MIT License
  - All models, datasets, schemas, and endpoints are 100% publicly accessible on GitHub.